

Technical Development Guide

Development Standards

Code Quality

All code must pass code review before deployment

Unit test coverage should be at least 80%

Follow established naming conventions and patterns

Security Protocols

Regular security audits quarterly

All APIs must implement rate limiting

Sensitive data must be encrypted at rest and in transit

Deployment Process

Staging Environment

All features must be tested in staging for 48 hours

Performance testing required for high-traffic features

Database migrations must be backward compatible

Production Deployment

Deployments occur every Tuesday and Thursday

Rollback plan must be documented for each deployment

Monitoring alerts must be configured for new services

API Documentation Standards

All endpoints must have OpenAPI documentation

Include request/response examples

Document error codes and troubleshooting steps